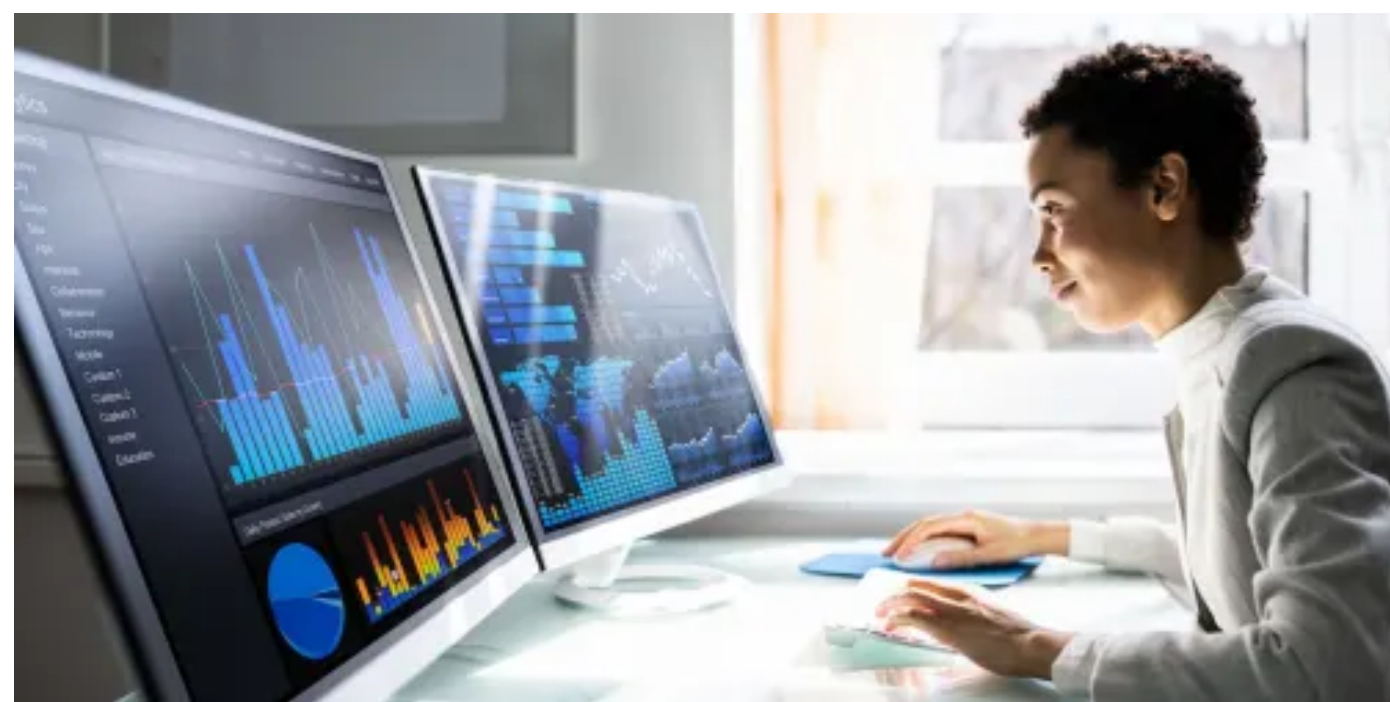


4 TYPES OF DATA ANALYTICS TO IMPROVE DECISION-MAKING



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Data is a powerful tool that's available to organizations at a staggering scale. When harnessed correctly, it has the potential to drive [decision-making](#), impact [strategy formulation](#), and improve organizational performance.

According to [The Global State of Enterprise Analytics](#) (pdf) report by business intelligence company MicroStrategy, 56 percent of respondents said data analytics led to “faster, more effective decision-making” at their companies. Other benefits cited include:

- Improved efficiency and productivity (64 percent)
- Better financial performance (51 percent)
- Identification and creation of new product and service revenue (46 percent)
- Improved customer acquisition and retention (46 percent)
- Improved customer experiences (44 percent)
- Competitive advantage (43 percent)

How can you harness the power of data and experience these benefits at your company? Learning how to analyze data effectively can enable you to draw conclusions, predictions, and actionable insights to drive impactful decision-making.

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WHAT IS DATA ANALYTICS IN BUSINESS?

Data analytics is the practice of examining data to answer questions, identify trends, and extract insights. When data analytics is used in business, it's often called [business analytics](#).

You can use tools, frameworks, and software to analyze data, such as Microsoft Excel and Power BI, Google Charts, Data Wrapper, Infogram, Tableau, and Zoho Analytics. These can help you examine data from different angles and [create visualizations](#) that illuminate the story you're trying to tell.

Algorithms and machine learning also fall into the data analytics field and can be used to gather, sort, and analyze data at a higher volume and faster pace than humans can. Writing algorithms is a more advanced data analytics skill, but you don't need deep knowledge of coding and statistical modeling to experience the benefits of data-driven decision-making.

Related: [The Advantages of Data-Driven Decision-Making](#)

WHO NEEDS DATA ANALYTICS?

Any business professional who makes decisions needs foundational data analytics knowledge. Access to data is more common than ever. If you formulate strategies and make decisions without considering the data you have access to, you could miss major opportunities or red flags that it communicates.

Professionals who can benefit from data analytics skills include:

- Marketers, who utilize customer data, industry trends, and performance data from past campaigns to plan marketing strategies
- Product managers, who analyze market, industry, and user data to improve their companies' products
- Finance professionals, who use historical performance data and industry trends to forecast their companies' financial trajectories
- Human resources and diversity, equity, and inclusion professionals, who gain insights into employees' opinions, motivations, and behaviors and pair it with industry trend data to make meaningful changes within their organizations

To get the greatest insight from your data, familiarize yourself with the four key types of data analytics. Here's a breakdown of the types, which you can use individually or in tandem to maximally benefit from your company's data.

Check out our video on business analytics below, and subscribe to our [YouTube channel](#) for more explainer content!

What Is

Business Analytics?

4 KEY TYPES OF DATA ANALYTICS

1. Descriptive Analytics

[Descriptive analytics](#) is the simplest type of analytics and the foundation the other types are built on. It allows you to pull trends from raw data and succinctly describe what happened or is currently happening.

Descriptive analytics answers the question, “What happened?”

For example, imagine you’re analyzing your company’s data and find there’s a seasonal surge in sales for one of your products: a video game console. Here, descriptive analytics can tell you, “This video game console experiences an increase in sales in October, November, and early December each year.”

[Data visualization](#) is a natural fit for communicating descriptive analysis because charts, graphs, and maps can show trends in data—as well as dips and spikes—in a clear, easily understandable way.

2. Diagnostic Analytics

[Diagnostic analytics](#) addresses the next logical question, “Why did this happen?”

Taking the analysis a step further, this type includes comparing coexisting trends or movement, uncovering correlations between variables, and determining causal relationships where possible.

Continuing the aforementioned example, you may dig into video game console users' demographic data and find that they're between the ages of eight and 18. The customers, however, tend to be between the ages of 35 and 55. Analysis of customer survey data reveals that one primary motivator for customers to purchase the video game console is to gift it to their children. The spike in sales in the fall and early winter months may be due to the holidays that include gift-giving.

Diagnostic analytics is useful for getting at the root of an organizational issue.

3. Predictive Analytics

[Predictive analytics](#) is used to make predictions about future trends or events and answers the question, "What might happen in the future?"

By analyzing historical data in tandem with industry trends, you can make informed predictions about what the future could hold for your company.

For instance, knowing that video game console sales have spiked in October, November, and early December every year for the past decade provides you with ample data to predict that the same trend will occur next year. Backed by upward trends in the video game industry as a whole, this is a reasonable prediction to make.

Making predictions for the future can help your organization formulate strategies based on likely scenarios.

4. Prescriptive Analytics

Finally, [prescriptive analytics](#) answers the question, "What should we do next?"

Prescriptive analytics takes into account all possible factors in a scenario and suggests actionable takeaways. This type of analytics can be especially useful when making data-driven decisions.

Rounding out the video game example: What should your team decide to do given the predicted trend in seasonality due to winter gift-giving? Perhaps you decide to run an [A/B test](#) with two ads: one that caters to product end-users (children) and one targeted to customers (their parents). The data from that test can inform how to capitalize on the seasonal spike and its supposed cause even further. Or, maybe you decide to increase marketing efforts in September with holiday-themed messaging to try to extend the spike into another month.

While manual prescriptive analysis is doable and accessible, machine-learning algorithms are often employed to help parse through large volumes of data to recommend the optimal next step. Algorithms use "if" and "else" statements, which work as rules for parsing data. If a specific combination of requirements is met, an algorithm recommends a specific course of action. While there's far more to machine-learning algorithms than just those statements, they—along with mathematical equations—serve as a core component in algorithm training.



USING DATA TO DRIVE DECISION-MAKING

The four types of data analysis should be used in tandem to create a full picture of the story data tells and make informed decisions. To understand your company's current situation, use descriptive analytics. To figure out how your company got there, leverage diagnostic analytics. Predictive analytics is useful for determining the trajectory of a situation—will current trends continue? Finally, prescriptive analytics can help you consider all aspects of current and future scenarios and plan actionable strategies.

Depending on the problem you're trying to solve and your goals, you may opt to use two or three of these analytics types—or use them all in sequential order to gain the deepest understanding of the story data tells.

Strengthening your [analytics skills](#) can empower you to take advantage of insights your data offers and advance your organization and career.